Multi-Omics Target Prioritization in Ovarian Cancer

1. Disease & Data Choice

I chose high-grade serous ovarian cancer (HGSOC) because it offered genuinely sample-matched, three-layer proteogenomic data through the TCGA/CPTAC ovarian cancer study (Zhang et al., 2016). All three data layers came from the same underlying cohort (TCGA/CPTAC ovarian serous cystadenocarcinoma), distributed through LinkedOmics as pre-processed matrices (Vasaikar et al., 2017):

Layer	Source / File	Samples	Genes	Access
Genomic	LinkedOmics (WUSM) - Mutation, gene level, MutSig2CV-filtered (.cbt)	465 tumors	2,205	Open
Transcriptomic	LinkedOmics (UNC) - RNAseq, Illumina HiSeq, RSEM log2 (.cct)	303 tumors	20,032	Open
Proteomic	LinkedOmics (PNNL) - Proteome, iTRAQ log-ratio, gene level (.cct)	84 tumors	6,475	Open

All three files were open-access, no data-use agreement required. The mutation matrix was reduced from a gene-by-tumor binary matrix to per-gene mutation frequency. RNA and protein matrices were likewise reduced to gene-level evidence for scoring. Importantly, 54 of the 84 proteome-profiled tumors also had RNA-seq measurements, allowing sample-level RNA-protein concordance to be assessed in the same tumors.

2. Multi-Omics Integration and Scoring

Gene identifiers were harmonized across layers, and the matrices were joined on gene symbol. Because the mutation data were provided as a binary gene-by-tumor matrix, they were first reduced to per-gene mutation frequency across the 465 tumors. The three-way join retained 751 genes from 20,032 RNA-seq genes, 6,475 proteomic genes, and 2,205 mutation-tested genes. Because the public RNA/protein matrices do not contain tumor/normal labels, their gene-level evidence represents overall abundance magnitude rather than tumor-vs-normal fold change.

For concordance, a signed fold-change comparison like the Alzheimer's exercise was not appropriate. Instead, the matched design allowed a per-gene Spearman correlation between RNA and protein across the 54 shared tumors. Median RNA-protein correlation across the 751 joined genes was 0.49 (range -0.18 to 0.88), showing that RNA and protein are related but far from interchangeable.

For prioritization, each layer was converted to a 0-1 percentile rank so that measurements on different scales could be combined. The three percentile ranks were then combined into a single multi-evidence score; the weighting strategy is discussed below.

3. Weighting

I used equal weights (1/3 each) for the transcriptomic, proteomic, and genomic layers because each captures a distinct aspect of tumor biology, and this dataset provides no systematic basis for prioritizing one layer over another. Up-weighting genomic evidence would also be defensible because recurrent mutation may be closer to a causal event than RNA or protein abundance. For example, TP53 ranks 9th of 751 largely because of its 82.8% mutation frequency, while its RNA and protein abundance are relatively unremarkable. However, adjusting the weights based on a single expected gene would introduce a post hoc bias. I therefore retained equal weighting as the most neutral and reproducible default.

4. Top Targets and Biological Interpretation

The ranked list of all 751 genes is provided in targets_ov.csv. The top 15 were:

Gene	Transcriptomic	Proteomic	Genomic	RNA-Prot r	Score
MUC16	13.564	0.476	0.043	0.664	0.988
COL6A3	12.327	0.196	0.030	0.343	0.951
FN1	14.175	0.409	0.011	0.714	0.913

MACF1	11.894	0.125	0.030	0.428	0.904
COL3A1	14.059	0.259	0.011	0.458	0.904
CHD4	12.674	0.107	0.015	0.088	0.889
MXRA5	11.579	0.306	0.013	0.613	0.888
KDM5C	11.520	0.149	0.017	n/a	0.884
TP53	11.023	0.151	0.828	n/a	0.875
PRKDC	12.189	0.099	0.015	0.295	0.867
COL1A2	14.837	0.389	0.009	0.297	0.858
NOTCH2	12.648	0.116	0.011	n/a	0.847
MYH9	14.285	0.087	0.011	0.661	0.829
BPTF	10.742	0.165	0.013	0.237	0.814
AHNAK	13.883	0.048	0.028	0.566	0.813

Known HGSOC biology was recovered in complementary ways. MUC16 ranked first (score 0.988); it encodes CA-125 and was driven mainly by high transcriptomic and proteomic abundance rather than mutation. TP53 ranked 9th (score 0.875) with the opposite profile: its 82.8% mutation frequency supplied most of its evidence, consistent with TP53 being a defining recurrent alteration in HGSOC (Bell et al, 2011; Zhang et al., 2016). RB1 (43/751), KRAS (41/751), and NF1 (53/751) also appeared within roughly the top 7% of the full list.

Not every known ovarian cancer gene could be recovered. BRCA1, BRCA2, MYC, and CCNE1 were absent from the 751-gene join because the 6,475-gene proteomic panel did not contain them. Their absence is therefore a coverage artifact rather than evidence against their biological relevance.

A non-obvious pattern was the visibility of extracellular-matrix genes, including COL6A3, COL3A1, COL1A2, FN1, and MXRA5. Their generally positive RNA-protein correlations

(approximately 0.29-0.71) suggest a coordinated signal. In bulk HGSOC tissue, this signal could reflect the stroma-rich tumor microenvironment rather than tumor-cell-intrinsic driver activity. Therefore, these hits are biologically interesting but should not automatically be interpreted as therapeutic targets.

5. Discordance and Limitations

Because tumor/normal fold changes were unavailable, I assessed RNA–protein concordance using Spearman correlations across 54 matched tumors. Positive correlations were classified as concordant and negative correlations as discordant, with undefined values left unclassified. This is a correlation-based proxy rather than a formal direction-of-change comparison. CPN2 showed the lowest correlation ($r = -0.18$) and relatively low multi-evidence score (0.375), despite high RNA abundance. This discrepancy could reflect post-transcriptional regulation, protein turnover, measurement variability, or sample composition, but the weak correlation does not establish a specific mechanism.

Several limitations also affect interpretation of the ranking. First, the RNA and protein datasets lack a tumor/normal comparator, so their evidence reflects overall abundance rather than cancer-specific differential expression. Second, proteomic coverage was the main bottleneck: only 6,475 of 20,032 RNA-seq genes had protein measurements, excluding known ovarian cancer genes such as BRCA1, BRCA2, MYC, and CCNE1 from the final three-way join. Third, the mutation dataset was pre-filtered by MutSig2CV to 2,205 genes, potentially limiting genomic evidence for less recurrently altered genes. Finally, because these are bulk tumor measurements, stromal and tumor-cell signals cannot be separated, which is particularly relevant to the ECM-rich top hits discussed above.

Despite these limitations, the 54 matched tumors allow sample-level RNA-protein concordance to be evaluated directly, an advantage over unmatched cross-cohort integration.

6. References

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